

Nathan J. Szymanski

ASSISTANT PROFESSOR · MATERIALS SCIENCE AND ENGINEERING

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Research Interests

I am a materials scientist working to advance the development of sustainable energy technologies by integrating computational and experimental methods. My work combines *ab-initio* quantum chemistry, machine learning, and Monte Carlo simulations to model inorganic materials synthesis. By pairing these techniques with *in-situ* characterization and AI-driven data analysis, my group streamlines the process between initial discovery and commercialization of next-generation materials for energy storage and high-temperature applications.

Education

University of California, Berkeley

Berkeley, CA

PHD MATERIALS SCIENCE AND ENGINEERING

2019 - 2024

University of Toledo

Toledo, OH

BS PHYSICS AND APPLIED MATHEMATICS

2015 - 2019

Professional Experience

- 2025- **Assistant Professor**, Materials Science and Engineering, University of California, Los Angeles
- 2024-2025 **Postdoctoral Researcher**, Chemical Engineering and Materials Science, University of Minnesota
- 2019-2024 **Graduate Research Assistant**, Materials Science and Engineering, University of California, Berkeley
- 2023-2024 **Graduate Teaching Assistant**, Graduate-level Thermodynamics, University of California, Berkeley

Publications

1. N. J. Szymanski, K. J. Warren, A. W. Weimer, & C. J. Bartel, Cation Vacancies Mediate Thermochemical Water Splitting with Iron Aluminates. *Chemistry of Materials* 38, 3145–3156 (2026).
2. M. Abolhasani, M. Ahmadi, S. Baird, C. P. Berlinguette, K. A. Brown, D. Fenning, I. T. Foster, W. Gottstein, S. Pablo-Garcia, G. Gomes, J. R. Hattrick-Simpers, J. E. Hein, T. Hitosugi, S. V. Kalinin, B. Maruyama, J. Schrier, M. Seifrid, T. D. Sparks, S. Sun, B. R. Sutherland, **N. J. Szymanski**, D. Thomas, L. Walters, J. Xu, Y. Zeng, C. Yang, On the Need for Autonomous Science Instruments: A Call to Action. *ChemRxiv* (2026).
3. J. Schlesinger, S. Hjaltason, **N. J. Szymanski**, & C. J. Bartel, Thermodynamic assessment of machine learning models for solid-state synthesis prediction. *arXiv:2602.04075* (2026).
4. A. J. Lannerd, **N. J. Szymanski**, and C. J. Bartel, Thermodynamics of proton insertion across the perovskite–brownmillerite transition in $\text{La}_{0.5}\text{Sr}_{0.5}\text{CoO}_{3-\delta}$. *Phys. Rev. Materials* 10, 025401 (2026).
5. L. Rettenberger, **N. J. Szymanski**, A. Giunto, O. Dartsis, A. Jain, G. Ceder, V. Hagenmeyer, and M. Reischl, Leveraging unlabeled SEM datasets with self-supervised learning for enhanced particle segmentation. *npj Computational Materials* 11, 289 (2025).
6. **N. J. Szymanski** and C. J. Bartel, Establishing baselines for the generative discovery of inorganic crystals, *Materials Horizons, Advance Article* (2025).

7. **N. J. Szymanski**, A. Smith, P. Daoutidis, and C. J. Bartel, Topological descriptors for the electron density of inorganic solids. *ACS Materials Letters* **7**, 2158–2164 (2025).
8. H. Kim, K. Jun, **N. J. Szymanski**, V. Sai, Z. Cai, M. Crafton, G.-H. Lee, S. E. Trask, F. Babbe, Y.-W. Byeon, P. Zhong, D. Lee, B. Park, W. Jung, B. D. McCloskey, and W. Yang, Screening and Development of Sacrificial Cathode Additives for Lithium-Ion Batteries, *Advanced Energy Materials*, 2403946 (2025).
9. V. S. Avvaru, T. Li, G.-H. Lee, Y.-W. Byeon, K. P. Koirala, O. J. Marques, S. Jeong, **N. J. Szymanski**, M. Kunz, F. Babbe, V. Battaglia, B. D. McCloskey, J. N. Weker, C. Wang, W. Yang, R. J. Clément, and H. Kim, Alternative Solid State Synthesis Route for Highly Fluorinated Disordered Rocksalt Cathode Materials for High Energy Lithium Ion Batteries, *Advanced Energy Materials*, 2500492 (2025).
10. Y. Chen, X. Zhao, K. Chen, K. P. Koirala, R. Giovine, X. Yang, S. Wang, **N. J. Szymanski**, S. Xiong, Z. Lun, H. Ji, C. Wang, J. Bai, F. Wang, B. Ouyang, G. Ceder, Coherent-Precipitation-Stabilized Phase Formation in Over-Stoichiometric Rocksalt-Type Li Superionic Conductors, *Advanced Materials*, 2416342 (2024).
11. **N. J. Szymanski** Y.-W. Byeon, Y. Sun, Y. Zeng, J. Bai, M. Kunz, C. J. Bartel, H. Kim, and G. Ceder, Quantifying the regime of thermodynamic control in solid-state reactions. *Science Advances* **10**, eadp3309 (2024).
12. **N. J. Szymanski** & C. J. Bartel, Computationally Guided Synthesis of Battery Materials. *ACS Energy Letters* **9**, 2902 (2024).
13. Y. Fei, B. Rendy, R. Kumar, O. Dartsy, H. P. Sahasrabudhe, M. J. McDermott, Z. Wang, **N. J. Szymanski**, L. N. Walters, D. Milsted, Y. Zeng, A. Jain, G. Ceder, AlabOS: a Python-based reconfigurable workflow management framework for autonomous laboratories. *Digital Discovery*, Advance Article (2024).
14. S. Wang, **N. J. Szymanski**, Y. Fei, Y. Zeng, M. Whittaker, & G. Ceder, Direct Lithium Extraction from α -Spodumene through Solid-State Reactions for Sustainable Li_2CO_3 Production. *Inorganic Chemistry* (2024).
15. **N. J. Szymanski**, S. Fu, E. Persson, & G. Ceder, Integrated analysis of X-ray diffraction patterns and pair distribution functions for machine-learned phase identification. *npj Computational Materials* **10**, 45 (2024).
16. Y. Zeng* and **N. J. Szymanski***, T. He, K. Jun, L. C. Gallington, H. Huo, C. J. Bartel, B. Ouyang, & G. Ceder, Selective formation of metastable polymorphs in solid-state synthesis. *Science Advances* **10**, eadj5431 (2024).
17. L. Rettenberger, **N. J. Szymanski**, Y. Zeng, J. Scheutcke, S. Wang, G. Ceder, & M. Reischl, Uncertainty-aware particle segmentation for electron microscopy at varied length scales. *npj Computational Materials* **10**, 124 (2024).
18. **N. J. Szymanski***, B. Rendy*, Y. Fei*, R. E. Kumar*, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng, & G. Ceder, An autonomous laboratory for the accelerated synthesis of novel materials. *Nature* **624**, 86-91 (2023).
19. **N. J. Szymanski***, P. Nevatia*, C. J. Bartel, Y. Zeng, & G. Ceder, Autonomous and dynamic precursor selection for solid-state materials synthesis. *Nature Communications* **14**, 6956 (2023).
20. **N. J. Szymanski**, C. J. Bartel, Y. Zeng, M. Diallo, H. Kim, & G. Ceder, Adaptively driven X-ray diffraction guided by machine learning for autonomous phase identification. *npj Computational Materials* **9**, 31 (2023).
21. **N. J. Szymanski**, S. Fu, E. Persson, & G. Ceder, Z. Lun, J. Liu, E. C. Self, C. J. Bartel, J. Nanda, B. Ouyang, & G. Ceder, Modeling Short-Range Order in Disordered Rocksalt Cathodes by Pair Distribution Function Analysis. *Chemistry of Materials* **35**, 4922-4934 (2023).
22. Y. W. Byeon, M. J. Gong, Z. Cai, Y. Sun, **N. J. Szymanski**, J. Bai, D. H. Seo, & H. Kim, Effects of cation and anion substitution in KVPO_4F for K-ion batteries. *Energy Storage Materials* **57**, 81-91 (2023).
23. **N. J. Szymanski**, Y. Zeng, T. Bennett, S. Patil, J. K. Keum, E. C. Self, J. Bai, Z. Cai, R. Giovine, B. Ouyang, F. Wang, C. J. Bartel, R. J. Clément, W. Tong, J. Nanda, & G. Ceder, Understanding the Fluorination of Disordered Rocksalt Cathodes through Rational Exploration of Synthesis Pathways. *Chemistry of Materials* **34**, 7015-7028 (2022).

24. **N. J. Szymanski**, Y. Zeng, H. Huo, C. J. Bartel, H. Kim, & G. Ceder, Toward autonomous design and synthesis of novel inorganic materials. *Materials Horizons* 8, 2169-2198 (2021).
25. **N. J. Szymanski**, C. J. Bartel, Y. Zeng, Q. Tu, & G. Ceder, Probabilistic deep learning approach to automate the interpretation of multi-phase diffraction spectra. *Chemistry of Materials* 33, 4204-4215 (2021).
26. I. Khare, **N. J. Szymanski**, D. Gall, & R. Irving, Electronic, optical, and thermoelectric properties of sodium pnictogen chalcogenides: A first principles study. *Computational Materials Science* 183, 109818 (2020).
27. **N. J. Szymanski***, L. N. Walters*, D. Puggionio, & J. Rondinelli, Design of heteroanionic MoON exhibiting a Peierls metal-insulator transition. *Physical Review Letters* 123, 236402 (2019).
28. **N. J. Szymanski**, V. Adhikari, M. A. Willard, P. Sarin, D. Gall, & S. V. Khare, Prediction of improved magnetization and stability in Fe₁₆N₂ through alloying. *Journal of Applied Physics* 127, 093903 (2019).
29. **N. J. Szymanski**, I. Khatri, J. G. Amar, D. Gall, & S. V. Khare, Unconventional superconductivity in 3d rocksalt transition metal carbides. *Journal of Materials Chemistry C* 7, 12619 (2019).
30. **N. J. Szymanski**, V. Adhikari, I. Khatri, D. Gall, & S. V. Khare, Improved optoelectronic properties in CdSe_xTe_{1-x} through controlled composition and short-range order. *Solar Energy* 194, 742 (2019).
31. V. Adhikari, **N. J. Szymanski**, I. Khatri, D. Gall, & S. V. Khare, First principles investigation into the phase stability and enhanced hardness of TiN-ScN and TiN-YN alloys. *Thin Solid Films* 688, 137284 (2019).
32. **N. J. Szymanski**, L. N. Walters, O. Hellman, D. Gall, & S. V. Khare, Dynamical stabilization in delafossite nitrides for solar energy conversion. *Journal of Materials Chemistry A* 6, 20852 (2018).
33. **N. J. Szymanski**, Z. T. Y. Liu, T. Alderson, N. J. Podraza, P. Sarin, & S. V. Khare, Electronic and optical properties of vanadium oxides from first principles. *Computational Materials Science* 146, 310 (2018).
34. V. Adhikari, Z. T. Y. Liu, **N. J. Szymanski**, I. Khatri, D. Gall, P. Sarin, & S. V. Khare, First-principles study of mechanical and magnetic properties of transition metal (M) nitrides in the cubic M₄N structure. *Journal of Physics and Chemistry of Solids* 120C, 197 (2018).

Patents Pending

- 2024 **Li extraction from spodumene**, A method of synthesizing Li precursors directly from hard minerals.

Awards & Fellowships

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| 2024 | Didier de Fontaine Award in Theory and Computation , UC Berkeley | \$ 2,000 |
| 2023 | Jane Lewis Fellowship for Research in Mining , UC Berkeley | \$ 40,000 |
| | Vedensky Award in Process Metallurgy , UC Berkeley | \$ 1,000 |
| 2019 | NSF Graduate Research Fellowship , National Science Foundation | \$ 108,000 |

Software

CrysTopo. Tools for the topological analysis of DFT-computed charge densities.

MatGen-Baselines. A package to benchmark generative AI for materials discovery.

XRD-AutoAnalyzer. A package to automate phase identification from multi-phase XRD patterns.

ARROWS. A decision-making algorithm designed to optimize solid-state synthesis procedures.

AdaptiveXRD. Software that integrates automated XRD analysis with in-line steering of measurements.

Teaching Experience

- 2026 **COM SCI M148: Introduction to Data Science**, Instructor
- 2026 **MAT SCI 104: Introduction to Engineering Materials**, Instructor
- 2026 **MAT SCI 90L: Physical Measurements of Materials**, Instructor
- 2024 **Machine learning for Chemical Engineering and Materials Science**, Guest Lecturer
- 2023 **Thermodynamics and Phase Transformations**, Graduate Teaching Assistant